Alberto Cereser

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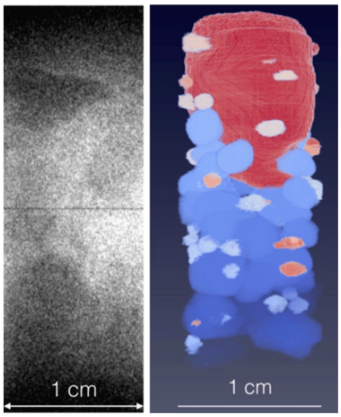

Summary

Physicist specialized in developing algorithms for image processing and data analysis. I contributed to the development of two new microscopy techniques. In the figure, typical input image (*left*) and final output (*right*) of the new microscopy technique I developed during my Ph.D. to study the internal structure of materials.

Education

2013 - 2016, Technical University of Denmark and European Spallation Source (ESS ERIC)

PhD in Physics. Project: *"Time-of-flight 3D Neutron Diffraction for Multigrain Crystallography"*

Supervisors: Dr. Søren Schmidt (main), Prof. Dr. Markus Strobl, Dr. Stephen Hall, Dr. Axel Steuwer

I developed a new microscopy technique to reconstruct, non-destructively, the 3D shape and crystallographic orientation of the grains composing crystalline materials (i.e. metals, rocks, . . .)

Tasks:

- Plan and coordinate sample preparation
- Simulate the experiments using McStas
- Perform experiments at neutron facilities worldwide. The experiments were the first of their kind, and required a new detector setup
- Develop ad-hoc image and data analysis algorithms to analyse the collected data
- Liaise between a team of 20+

2009 - 2011, Niels Bohr Institute, University of Copenhagen

M.Sc. in Physics. Thesis: *"Nanofocused X-ray Analysis of Semiconductor Nanowires"*

Supervisor: Prof. Dr. Robert Feidenhans'l

2004-2009, University of Padova

B.Sc. in Physics. Thesis: *"Analysis of Diffusion in Egg White by Fluorescence Correlation Spectroscopy"*

Supervisors: Prof. Dr. Thomas Heimburg, Prof. Dr. Matteo Pierno

Research experience

05/2017 - current, Technical University of Denmark

Postdoc.

Dark-field X-ray microscopy (DFXRM) is a new technique to reconstruct the 3D shape and orientation of the domains composing deeply embedded crystallites with a resolution of 100 nm. I am responsible for developing the reconstruction software, returning 3D shapes and crystallographic orientations.

Supervisor: Prof. Dr. Tech. Henning Friis Poulsen

02/2015 - 04/2015, Japan Proton Accelerator Research Complex (J-PARC)

Visiting student

11/2011 - 12/2012, ARC Center of Excellence for Coherent X-ray Science, Melbourne

Research assistant. Project: *"Metals in Alzheimer's Disease by Combined Synchrotron-based Approaches"*.

Other experience

2010 - 2014 The University Post, University of Copenhagen

Reporter. I wrote about 50 articles, mainly on research and development topics.

Computer skills

MATLAB, Python, R, C++, shell scripting, high performance computing (HPC), parallel computing

Competences

- Performing experiments at large-scale research facilities. Neutron techniques: imaging and scattering; X-ray techniques: imaging, scattering and fluorescence
- Synchrotron user experience at the European Synchrotron Light Source (ESRF, France), the Australian Synchrotron and the Advanced Photon Source (APS, USA)
- Experience with neutron experiments at ISIS (UK), PSI (Switzerland) and J-PARC (Japan)
- Sample preparation for single-cell X-ray imaging
- Diffusion measurements using fluorescence correlation spectroscopy (FCS)

Publications

Journal Articles

A. Cereser, M. Strobl, S. Hall, A. Steuwer, R. Kiyanagi, A. Tremsin, E. Bergbäck Knudsen, T. Shinohara, P. Willendrup, A. Bastos da Silva Fanta, S. Iyengar, P. M. Larsen, T. Hanashima, T. Moyoshi, P. M. Kadletz, P. Krooß, T. Niendorf, M. Sales, W. W. Schmahl & S. Schmidt. *Time-of-Flight Three Dimensional Neutron Diffraction in Transmission Mode for Mapping Crystal Grain Structures.* Nature Scientific Reports 7(1): 9561 (2017). DOI: 10.1038/s41598-017-09717-w.

PhD thesis

A. Cereser. *Time-of-flight 3D Neutron Diffraction for Multigrain Crystallography*. PhD thesis. Technical University of Denmark (2016).

Grants and funding

- During my PhD, I successfully applied for experiments at large-scale neutron facilities in Europe and Japan for a total of two months
- 2012 - Top-up scholarship, Commonwealth Scientific and Industrial Research Organisation (CSIRO)
- 2008 - Erasmus travel scholarship, European Commission

Referees

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